

Order-Flow Imbalance and Short-Horizon Return Predictability in Cryptocurrency Markets

A Review of the Evidence and a Reproducible Evaluation Framework

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Abstract

Order-flow imbalance, the net pressure of buyer- versus seller-initiated trading, is one of the most robust drivers of short-horizon price movement in market microstructure theory and equity evidence. This article consolidates that literature and translates it to cryptocurrency markets, where the aggressor side of each trade is reported rather than estimated and where tick-level data are freely public. We review the theoretical foundations and the empirical record on order-flow imbalance, define the principal measures precisely, describe the public data environment, and set out a transparent, reproducible framework for evaluating whether order flow *predicts* returns, as distinct from the largely mechanical relation by which trades move price within the same interval. The framework specifies out-of-sample evaluation, a realistic transaction-cost model, and explicit control for data-snooping. **This is a review and design article; it does not report new empirical estimates.** Its contribution is a consolidated account of what is known and a rigorous, open-data plan for testing it.

Keywords: market microstructure; order-flow imbalance; cryptocurrency; return predictability; high-frequency data; reproducibility.

JEL classification: G12; G14; G15; C58.

1. Introduction

Market microstructure theory holds that prices respond to *order flow*, the net pressure of buyer- versus seller-initiated trading. Under asymmetric information, marketable orders convey information and force liquidity providers to revise quotes, so signed order flow becomes the primary channel through which information enters prices at short horizons (Kyle, 1985; Glosten and Milgrom, 1985; Hasbrouck, 1991). The most widely cited empirical sharpening of this idea is due to Cont, Kukanov, and Stoikov (2014), who show that over short intervals the change in price is an approximately linear function of **order-flow imbalance (OFI)**, with a slope that falls as market depth rises.

Cryptocurrency markets are a natural setting in which to revisit these results. Two features distinguish them from the equity markets in which OFI was first studied. First, the aggressor side of each trade is *observed*: exchange feeds report directly whether the buyer or the seller was the liquidity taker, so signed order flow is recovered exactly, without the trade-classification error that attends heuristics such as Lee and Ready (1991). Second, the data are public and granular, years of tick-level trades are freely downloadable, which makes complete, end-to-end reproduction feasible in a way that proprietary equity and futures datasets rarely allow.

This article has three aims. It **reviews** the theory and the empirical record linking order-flow imbalance to returns, in both equities and crypto. It **defines** the principal order-flow measures and the public data environment precisely enough to be implemented. And it **specifies a reproducible framework** for the central open question: does order flow observed up to a given moment *predict* subsequent returns, and is any such predictability robust and economically meaningful? The distinction between this predictive question and the largely mechanical *contemporaneous* relation, that trades move price within the same interval, is the conceptual spine of the article. The two are routinely conflated in practitioner writing; separating them is what allows order-flow evidence to bear on the informational efficiency of these markets.

To be explicit about scope: this is a review and design article and does not present new empirical estimates. Sections 2–4 synthesise existing theory and evidence; Section 5 defines the measures; Section 6 describes the data; Section 7 sets out the evaluation framework; Sections 8–9 discuss and conclude.

2. Theoretical Background

The link between order flow and price originates in models of trading under asymmetric information. Kyle (1985) shows that an informed trader’s order flow is impounded into price linearly, defining a depth parameter (the reciprocal of “Kyle’s lambda”) that governs the price impact of a given quantity. Glosten and Milgrom (1985) derive the bid–ask spread as compensation that liquidity providers require for the adverse-selection risk that some incoming orders are informed; each trade therefore shifts the market maker’s beliefs and the quotes. Hasbrouck (1991) operationalises these ideas with a vector-autoregressive treatment of signed trades and quote revisions, measuring the persistent price impact, the information content, of a trade. The common thread is that signed order flow, not price pattern, is the proximate cause of short-horizon price change, and that its impact scales inversely with available liquidity.

3. Evidence on Order-Flow Imbalance

Cont, Kukanov, and Stoikov (2014) provide the benchmark empirical result. Using exchange data for a cross-section of US equities, they document an approximately linear contemporaneous relation between best-level order-flow imbalance and mid-price changes, with a slope inversely proportional to depth; the relation is reported to be stable across stocks and time scales and more robust than the familiar square-root relation between price moves and traded volume. Xu, Gould, and Howison (2018) extend OFI to multiple levels of the limit order book and find that flow at deeper levels carries additional explanatory power. Cont, Cucuringu, and Zhang (2023) construct an integrated, cross-asset formulation that improves on the best-level measure and captures cross-impact between instruments.

Alongside these results, a methodological literature supplies the tools for evaluating predictive ability honestly once many predictors and models are entertained: White (2000) and Hansen (2005) for data-snooping-robust inference, Diebold and Mariano (1995) and Clark and West (2007) for forecast comparison, and Campbell and Thompson (2008) for out-of-sample assessment against a historical-mean benchmark. Table 1 summarises the principal strands of evidence.

Table 1. Selected evidence on order-flow imbalance and returns.

Study	Market / data	Measure	Principal finding
Cont, Kukanov & Stoikov (2014)	US equities (exchange TAQ)	Best-level OFI	Near-linear contemporaneous price impact; slope falls with depth
Xu, Gould & Howison (2018)	US equities (limit order book)	Multi-level OFI	Deeper-level flow adds explanatory power over best level
Cont, Cucuringu & Zhang (2023)	US equities	Integrated / cross-asset OFI	Cross-impact; integrated measure improves on best-level OFI
Easley, López de Prado & O'Hara (2012)	US futures	VPIN (flow toxicity)	Order-flow toxicity tracks liquidity provision and volatility
Silantyev (2019)	BitMEX XBTUSD perpetual	Trade-flow imbalance vs aggregate OFI	TFI explains contemporaneous price changes better than aggregate OFI
Recent crypto order-flow study (2026)	Cross-section of cryptocurrencies	International order flow	Conditioning on order flow improves out-of-sample return forecasts
Recent crypto LOB study (2026)	Binance perpetuals (BTC, ETH, +)	Engineered book/trade features	Stable cross-asset predictive importance; validated under SPA testing

4. Order Flow in Cryptocurrency Markets

Applications to digital assets are more recent but converge on the relevance of order flow. Silantyev (2019), studying the BitMEX perpetual, finds that a **trade-flow imbalance (TFI)** measure computed from signed taker volume explains contemporaneous price changes better than a book-based aggregate, a result that both motivates an emphasis on TFI and is convenient given the wide public availability of trade data. Subsequent work reports that conditioning on order flow improves out-of-sample machine-learning forecasts of cryptocurrency returns relative to fundamental predictors, and recent high-frequency studies on Binance perpetual contracts document that engineered order-book and trade features carry stable predictive importance across assets of very different capitalisation, validated with predictive-ability testing. A parallel line of work connects order-flow toxicity (the VPIN metric of Easley, López de Prado, and O’Hara, 2012) to Bitcoin volatility and price jumps.

Taken together, the evidence establishes a strong contemporaneous relation and suggestive predictive value, but leaves open precisely the questions this framework targets: how much predictive content survives *out-of-sample*, over which horizons, and whether it is exploitable once realistic costs and the multiplicity of tested specifications are accounted for.

5. Measuring Order Flow

Crypto trade records include a flag indicating whether the buyer was the maker, which maps directly to an aggressor sign: when the buyer is the maker the seller is the liquidity taker and the trade is seller-initiated (-1); otherwise it is buyer-initiated ($s = +1$). Signed trade volume is $s \cdot q$, with q the trade quantity. Because the aggressor is reported rather than inferred, no Lee–Ready classification is required and the corresponding measurement error is absent. Table 2 lists the measures used in the framework.

Table 2. Order-flow measures.

Measure	Definition	Role
TFI	Net signed taker volume in a bar: sum of $s \cdot q$ over trades	Primary contemporaneous and predictive regressor
TFI_norm	$(V_{\text{buy}} - V_{\text{sell}}) / (V_{\text{buy}} + V_{\text{sell}})$, bounded in $[-1, 1]$	Scale-free imbalance; cross-asset comparison
CVD	Running cumulative sum of signed volume	Regime and descriptive analysis
OFI	Best bid/ask size and price increments (book-based)	Robustness, where quote history is available

6. The Public Data Environment

Binance publishes a free market-data archive (`data.binance.vision`) of individual trades, aggregated trades, and candlesticks for spot pairs, with a public REST endpoint (`data-api.binance.vision`) for gap-filling; no account or API key is required. Aggregated-trade records carry the maker/taker flag needed to sign order flow. Tick data are large, several gigabytes per asset-year, so a practical pipeline processes data month-by-month and persists only bar-level features. Historical best-bid/ask (quote) records are available

only through approximately 2024, which is why a trade-based measure (TFI) is the natural primary object and book-based OFI is reserved for a bounded robustness check. Table 3 summarises the relevant sources.

Table 3. Public data sources.

Source	Content	Access
<code>data.binance.vision</code>	Bulk CSV: trades, aggregated trades, candlesticks	Free, no key; bulk download
<code>data-api.binance.vision</code>	REST endpoints for trades, aggTrades, depth, klines	Free, no key; rate-limited
Historical bookTicker dumps	Best bid/ask quotes (through ~2024)	Free; limited recency
Second venue (e.g., Coinbase)	Public trade data for external-validity check	Free / public

7. A Reproducible Evaluation Framework

The framework below specifies how the open questions *should* be tested. It is a design, not a set of results.

7.1 Separating contemporaneous from predictive content

The contemporaneous regression $r_t = \alpha + \beta \cdot \text{TFI}_t + \varepsilon_t$ (estimated on short subsamples and averaged, with Newey–West HAC standard errors) recovers the depth-dependent price-impact relation of Cont et al. (2014) in crypto and is largely mechanical. The economically meaningful object is the *predictive* regression of forward returns on *lagged* flow, $r_{\{t-t+h\}} = \alpha + \sum \beta_k \text{OF}_{\{t-k\}} + \gamma r_{\{t-1\}} + \varepsilon$, evaluated against autoregressive and historical-mean benchmarks across horizons h .

7.2 Out-of-sample protocol

Data are partitioned chronologically (estimation, validation for any tuning, and a final test window touched only once) to preclude look-ahead. Predictive ability is judged primarily by the Campbell–Thompson out-of-sample R^2 , with the Clark–West statistic for nested-model comparison and the Diebold–Mariano statistic where applicable. A bivariate vector autoregression in returns and flow, with Granger-causality tests and impulse responses, characterises lead–lag dynamics and horizon decay.

7.3 Economic significance and costs

Any signal is mapped to a fully specified rule (position in the sign of the predicted return above a validation-tuned threshold, flat otherwise) and evaluated net of a conservative taker fee (on the order of ten basis points per side), a half-spread, and slippage. The reported quantities include turnover, net Sharpe ratio, and the break-even cost at which net value reaches zero. The interpretive rule is fixed in advance: predictability whose break-even cost lies below realistic trading costs is judged statistically real but not exploitable at retail latency, a conclusion fully consistent with efficient-markets reasoning.

7.4 Guarding against data-snooping

Because such a study searches over assets, horizons, resolutions, and specifications, conventional p-values overstate significance. The framework evaluates the full set of candidate models jointly with White’s Reality Check and Hansen’s Superior Predictive Ability test and reports conclusions under family-wise error control. Robustness analyses split by volatility regime and by bull/bear subperiod, add intraday-

seasonality controls, examine microstructure noise, compare TFI with book-based OFI where quotes permit, and replicate on a second venue.

8. Discussion

The literature reviewed here makes two things clear. The contemporaneous association between order flow and price is strong and well established, in equities and, on the available crypto evidence, in digital assets too. But a strong contemporaneous relation says little about market efficiency, because it is partly definitional: trades move price as they execute. The open and economically interesting question is whether order flow forecasts returns beyond that mechanical link, and whether such forecasts withstand realistic costs and honest multiple-testing discipline. The crypto setting is unusually well suited to answering it precisely because the aggressor side is observed and the data are public.

Several caveats temper any expectation. Microstructure predictability, where it exists, tends to be concentrated at latencies and queue positions inaccessible to ordinary participants, so statistical significance need not imply tradability, a distinction the cost model is designed to surface. Findings are typically venue- and period-specific; regime splits and second-venue replication bound but do not remove this concern. And best-level measures discard information in deeper book levels, available only intermittently in free crypto data. These are reasons to evaluate carefully, not reasons the question is uninteresting.

9. Conclusion

Order-flow imbalance is among the most dependable short-horizon drivers of price in microstructure theory and equity evidence, and the emerging crypto literature suggests the relationship carries over. This article has consolidated that evidence, defined the measures and the public data environment, and set out a transparent, reproducible framework that separates the mechanical contemporaneous relation from genuine out-of-sample predictability and subjects the latter to realistic costs and data-snooping control. The framework is deliberately implementable on free, public data, so that the empirical questions it poses can be answered, by the author or by any independent reader, with full reproducibility. Executing that empirical programme is the natural next step.

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Note: the two 2026 crypto studies referenced in the text are recent and are cited descriptively; full bibliographic details and all citations should be verified against primary sources and formatted to the target outlet's style before any submission.